



Lecture-1 (What is Database)

What is the difference between data and information?

Data

- Raw facts
- Not processed
- Has no direct meaning

Example:

45, 60, 72(just numbers)

Information

- Processed data
- Has meaning
- Useful for decision-making

Example:

"The average score of the class is 59."

What is a database?

A **database** is an organized collection of data that can be **stored**, **managed**, and **retrieved** easily.

In a database:

1. Store Data

Save information in tables.

Example: Add a new student record.

2. Retrieve Data

Search and view data when needed.

Example: Find a student by ID.

3. Update Data

Modify existing information.

Example: Change a student's phone number.

4. Delete Data

Remove unwanted records.

Example: Remove a student who graduated.

5. Organize Data

Arrange data in tables, rows, columns.

6. Manage Access (Security)

Control who can view or change the data.

7. Maintain Data Integrity

Ensure data is correct, consistent, and accurate.

Why do we use a database instead of memory?

1. Memory is temporary

Data in RAM is lost when the computer turns off.

Database stores data permanently.

2. Memory is limited

RAM has small storage.

Database can store huge amounts of data.

3. Memory is not organized

RAM doesn't provide tables, searching, indexing.

Database organizes data properly.

4. Memory has no security

Anyone can access RAM data.

Databases provide passwords, roles, permissions.

5. Memory cannot handle many users

RAM can't manage multiple access safely.

Databases handle many users at the same time.

Easy Example

If you save your phone contacts in RAM:

- All contacts disappear when the phone restarts.
- No search, no backup, no security.

But a **database** keeps them safe, organized, and permanent.

What Is Abstraction in a Database?

Abstraction in a database means **hiding unnecessary details** and **showing only what the user needs**.

It makes the database easier to use.

Easy Real-Life Example

When you use a **bank ATM**:

- You only see buttons like *withdraw, balance, deposit*.
- You **do not** see how the bank stores your data, how queries run, or how money is processed.

This is **abstraction**.

Where Do We Use Databases?

We use databases almost everywhere where data needs to be stored safely and used later.

1. Banking

To store account details, transactions, ATM records.

2. Hospitals

To store patient information, reports, and appointments.

3. Schools & Universities

For student records, results, attendance.

4. E-commerce (Amazon, Daraz)

To manage products, orders, payments.

5. Social Media

Facebook, Instagram store posts, comments, messages.

6. Mobile Apps

Contacts, messages, call logs, gallery data.

7. Government Systems

National ID, passport, tax records.

8. Transportation

Airline ticket booking, bus seat reservation.

DDL vs DML

1. DDL (Data Definition Language)

Used to **define** or **change the structure** of the database.

Examples:

- `CREATE`
- `ALTER`
- `DROP`

Easy Example:

```
CREATE TABLE Students (id INT, name VARCHAR(50));
```

2. DML (Data Manipulation Language)

Used to **insert, delete, update, or retrieve** data from tables.

Examples:

- `INSERT`
- `UPDATE`
- `DELETE`
- `SELECT`

Easy Example:

```
INSERT INTO Students VALUES (1, 'Ali');
```

Difference Between DDL and DML

Feature	DDL	DML
Meaning	Defines database structure	Manages data inside tables

Feature	DDL	DML
Full Form	Data Definition Language	Data Manipulation Language
Affects	Schema (table design)	Data (records/rows)
Examples	CREATE, ALTER, DROP	SELECT, INSERT, UPDATE, DELETE
Auto-commit	Yes (changes cannot be undone)	No (can be rolled back)
Use Case	Creating or modifying tables	Adding or modifying data